Celestial Mechanics 1

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1 Newton's Law of Gravitation

Celestial mechanics explores the trajectories of celestial objects based on gravitational interactions. The foundation is Newton's law of gravitation:

$$F(r) = \frac{GMm}{r^2}. \tag{1}$$

It is valid for all spherically symmetric objects. Two things are verifiably true:

All spherically symmetric objects exert force as if their mass is concentrated at the center. Only the mass enclosed by a given surface of special symmetries (spherical, cylindrical) exerts gravitational field; mass outside the said surface has no contribution.
Now we arrive to the concept of flux, which is the dot product of the field passing through a given area, and the area itself it is written as

$$\iint_S \vec{g} \cdot d\vec{A}$$

which simply implies the sum of dot product of every infinitesimal part of any enclosed area (such as a sphere) and the gravitational field at that infinitesimal area. Thus, the gravitational flux is

$$\Phi_g = \iint_S \vec{g} \cdot d\vec{A} = g(r) \cdot A = \left(-\frac{GM}{r^2}\right) \cdot (4\pi r^2).$$

Simplifying,

$$\Phi_g = 4\pi GM.$$

Therefore, the flux is independent of the area of the enclosed surface or distance from the mass. Rather, it depends only on the enclosed mass. This identity is useful for any spherical and cylindrical symmetry, which for now we won't get into.

Problem

Question:

1. What is the total gravitational flux of the Solar System? Assume the mass of other celestial objects is negligible comapared to the sun

2 Angular momentum and Energy conservation

Angular momentum of an object subjected to central force is always conserved which is found by the following observation that Consider a particle of mass m in the gravitational field of a point mass M at the origin.
The gravitational force on the particle is

$$\vec{F} = -\frac{GMm}{r^2}\hat{r},$$

where $r = |\vec{r}|$ and $\hat{r} = \frac{\vec{r}}{r}$ is the radial unit vector.
The torque is defined as

$$\vec{\tau} = \vec{r} \times \vec{F}.$$

Substituting $\vec{F}$,

$$\vec{\tau} = \vec{r} \times \left(-\frac{GMm}{r^2}\hat{r}\right).$$

But $\hat{r}$ is parallel to $\vec{r}$, hence

$$\vec{r} \times \hat{r} = 0.$$

Therefore,

$$\vec{\tau} = 0.$$

Thus, the torque on the particle due to the central gravitational force is zero. This implies that the angular momentum $\vec{L} = \vec{r} \times m\vec{v}$ is conserved.
Similarly, energy is also conserved as it can be quickly found that

$$\nabla \times \vec{F} = \vec{0}, \quad (r \neq 0).$$

And hence the force is conservative and thus at two different points, 1 and 2, along its trajectory:

$$\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$$

energy is conserved.

Problem

Let there be a planet of mass m orbiting a star of mass $M_\odot$ in an **elliptical orbit**.
The orbital distances are:

- Periapsis: $r_p = 0.8$ AU
- Apoapsis: $r_a = 1.2$ AU

Tasks:

1. Find the difference between the velocities at apoapsis and periapsis, $v_a - v_p$.
2. Take the average of the velocities and distances and **approximate the orbital period** of the planet.

Constants and relations:

- Mass of the star: $M_\odot = 1.989 \times 10^{30}$ kg
- Gravitational constant: $G = 6.674 \times 10^{-11}$ Nm^2/kg^2
- 1 AU $= 1.496 \times 10^{11}$ m

Solution

Step 1: Angular momentum conservation
The angular momentum is conserved(angle between velocity vector and radius vector is 90 degrees at both times)

$$L = mr_p v_p = mr_a v_a \quad \Longrightarrow \quad v_a = \frac{r_p}{r_a} v_p.$$

Step 2: Energy conservation between aphelion and perihelion
The total mechanical energy is the same at both points:

$$\frac{1}{2}mv_p^2 - \frac{GM_\odot m}{r_p} = \frac{1}{2}mv_a^2 - \frac{GM_\odot m}{r_a}.$$

Substitute $v_a = \frac{r_p}{r_a} v_p$:

$$\frac{1}{2}v_p^2 - \frac{GM_\odot}{r_p} = \frac{1}{2}\left(\frac{r_p}{r_a} v_p\right)^2 - \frac{GM_\odot}{r_a}.$$

Step 3: Solve for v_p

$$\frac{1}{2}v_p^2 - \frac{GM_\odot}{r_p} = \frac{1}{2}\frac{r_p^2}{r_a^2} v_p^2 - \frac{GM_\odot}{r_a}$$

$$v_p^2\left(\frac{1}{2} - \frac{r_p^2}{2r_a^2}\right) = GM_\odot\left(\frac{1}{r_p} - \frac{1}{r_a}\right)$$

$$v_p^2 = \frac{2GM_\odot\left(\frac{1}{r_p} - \frac{1}{r_a}\right)}{1 - \frac{r_p^2}{r_a^2}}. \tag{2}$$

Step 4: Compute v_a

$$v_a = \frac{r_p}{r_a} v_p$$

Step 5: Difference of velocities

$$\Delta v = v_p - v_a$$

Step 6: Approximate orbital period using average velocity
Average distance: $\bar{r} = \frac{r_p + r_a}{2}$
Average velocity: $\bar{v} = \frac{v_p + v_a}{2}$
Approximate period:

$$T \approx \frac{2\pi\bar{r}}{\bar{v}}$$

3 Total Energies of Circular, Elliptical and Parabolic orbit

3.1 Circular orbit

$$F_{\text{gravity}} = F_{\text{centripetal}}$$

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \Rightarrow \quad v^2 = \frac{GM}{r}$$

The kinetic energy is

$$K = \tfrac{1}{2}mv^2 = \tfrac{1}{2}m\left(\frac{GM}{r}\right) = \frac{GMm}{2r}$$

Thus, the total mechanical energy is

$$E = K + U = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r}$$

3.2 Elliptical orbit

Now for an elliptical orbit, we define the semi-major axis a as

$$a = \frac{r_p + r_a}{2}$$

where

$$r_p = \text{distance at perigee}$$
$$r_a = \text{distance at apogee}$$

Using this we can easily find the total energy of an object if we only substitute equation(2) for velocity from the previous example in a total energy equation. Thus The total orbital energy at periapsis is thus,

$$\begin{aligned} E &= \tfrac{1}{2}mv_p^2 - \frac{GMm}{r_p} \\ &= \frac{GMm}{1 - \frac{r_p^2}{r_a^2}}\left(\frac{1}{r_p} - \frac{1}{r_a}\right) - \frac{GMm}{r_p} \\ &= -\frac{GMm}{r_a + r_p} \end{aligned} \tag{3}$$

the orbital energy for the ellipse is

$$E_{\text{ellipse}} = -\frac{GMm}{2a}.$$

3.3 Parabolic orbit

A **parabolic** orbit is an orbit with an eccentricity of 1. This info is of no use for now, but assume its such an elliptical orbit where the apopasis is towards infinity Thus the quantity semi-major axis tends to infinity, and as a result,

$$E_{\text{parabola=0}}$$
$$E = \tfrac{1}{2}mv^2 - \frac{GMm}{r} = 0$$

4 Escape Velocity

Escape velocity is the minimum velocity needed for a body to move infinitely far from a gravitational source. If the initial velocity of a body is exactly equal to the escape velocity, as it moves away from the center, the velocity decreases until, asymptotically, it tends to zero. Therefore, the final kinetic and potential energies are zero, hence the total energy is zero as well. Because energy is conserved:

$$-\frac{GmM}{R} + \frac{1}{2}mv_{\text{esc}}^2 = 0,$$

where $m \ll M$ and R is the initial distance between the body and the centre of attraction. Solving for v_{esc}, we find:

$$\boxed{v_{\text{esc}} = \sqrt{\frac{2GM}{R}}}.$$

An interesting observation is that we encounter the same equation for the total energy of a parabolic orbit as in the previous section. This reveals that when a body's initial velocity is equal to the escape velocity, its trajectory naturally follows a parabolic path.
If a body moves with a velocity less than the escape velocity, it remains gravitationally bound to the central mass ineither a circular or elliptical orbit. When its velocity is exactly equal to the escape velocity, the trajectory becomes parabolic. For velocities greater than the escape velocity, the path turns **hyperbolic**.

Problem

Estimate the radius of a planet such that a man can escape its gravitation by jumping vertically. Assume density of the planet and the Earth are same. (109 – T03 – A)

Solution

Let us assume that a normal man can jump vertically up to 50 cm on the Earth. This number is purely a rough estimate and one can choose any number between 10 cm to 50 cm. One may think that a typical person's feet can get more than 50 cm away from the ground, but it should be remembered that vertical shift in his/her centre of gravity will be much smaller.

$$v_{\text{jump}} = \sqrt{2gh}$$

$$v_{\text{escape}} = \sqrt{\frac{2GM}{R}}$$

$$\therefore v_{\text{jump}} = v_{\text{escape}}$$

$$\therefore 2gh = \frac{2GM}{R} = \frac{2G}{R}\frac{4\pi\rho R^3}{3}$$

$$R = \sqrt{\frac{3gh}{4\pi\rho G}}$$

$$= \sqrt{\frac{3 \times 9.81 \times 0.5}{4\pi \times 5515 \times 6.6726 \times 10^{-11}}}$$

$$= 1.78 \times 10^3 \,\text{m}$$

$$R \approx 2\,\text{km or smaller}$$